WORKED SOLUTION

# STEP III 1995, Question 10

Mechanics • Projectiles on Inclined Planes

**STEP III 1995, Question 10**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q10

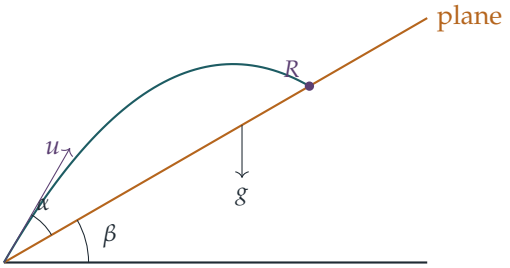
## The key idea

On an inclined plane the natural axes are not horizontal and vertical but *along* and *perpendicular to* the plane. In those axes gravity has two constant components, the time of flight comes from the perpendicular motion alone, and the range is a product of trigonometric functions. Turning that product into a difference of sines with the factor formula makes the maximum obvious, with no calculus needed.

## Solution

### Setting up

Model the cannon ball as a particle moving freely under gravity, with no air resistance, fired from the foot of the plane up its line of greatest slope. Let it be fired at angle $\alpha$ *above the plane*, so that its angle of elevation above the horizontal is $\theta = \alpha + \beta$; we need $0 < \alpha < \frac{\pi}{2} - \beta$ for the ball to go up the plane at all.




Take the $x$-axis up the line of greatest slope and the $y$-axis perpendicular to the plane, pointing away from it. Resolving the acceleration $g$ (vertically downwards) into these directions,

$$\ddot{x} = -g\sin\beta, \qquad \ddot{y} = -g\cos\beta,$$

and the initial velocity has components $u\cos\alpha$ along the plane and $u\sin\alpha$ perpendicular

to it. Hence at time $t$

$$x = ut\cos\alpha - \tfrac{1}{2}gt^2\sin\beta, \qquad y = ut\sin\alpha - \tfrac{1}{2}gt^2\cos\beta.$$

### Time of flight and range

The ball lands when $y = 0$ again with $t > 0$:

$$T = \frac{2u\sin\alpha}{g\cos\beta}.$$

The distance travelled up the plane is then

$$\begin{aligned} R = x(T) &= \frac{2u^2\sin\alpha\cos\alpha}{g\cos\beta} - \tfrac{1}{2}g\sin\beta\cdot\frac{4u^2\sin^2\alpha}{g^2\cos^2\beta} \\ &= \frac{2u^2\sin\alpha}{g\cos^2\beta}\big(\cos\alpha\cos\beta - \sin\alpha\sin\beta\big) \\ &= \frac{2u^2\sin\alpha\cos(\alpha+\beta)}{g\cos^2\beta}. \end{aligned}$$

**Ask yourself** How can we maximise $\sin\alpha\cos(\alpha+\beta)$ without differentiating?

The factor formula $2\sin A\cos B = \sin(A+B) + \sin(A-B)$, with $A = \alpha$ and $B = \alpha+\beta$, gives

$$2\sin\alpha\cos(\alpha+\beta) = \sin(2\alpha+\beta) - \sin\beta,$$

so

$$R = \frac{u^2\big(\sin(2\alpha+\beta) - \sin\beta\big)}{g\cos^2\beta}.$$

Only the term $\sin(2\alpha+\beta)$ depends on $\alpha$, and it is at most 1. As $\alpha$ runs over $(0, \frac{\pi}{2}-\beta)$, the angle $2\alpha+\beta$ runs over $(\beta, \pi-\beta)$, which contains $\frac{\pi}{2}$; so the maximum value 1 is attained, exactly when $2\alpha+\beta = \frac{\pi}{2}$, that is,

$$\alpha = \frac{\pi}{4} - \frac{\beta}{2}, \qquad \text{equivalently} \qquad \theta = \alpha+\beta = \frac{\pi}{4} + \frac{\beta}{2}.$$

So the cannon should be fired at $\frac{\pi}{4} + \frac{\beta}{2}$ above the horizontal: the firing direction bisects the angle between the plane and the vertical.

### The maximum range

With $\sin(2\alpha+\beta) = 1$, and using $\cos^2\beta = 1 - \sin^2\beta = (1-\sin\beta)(1+\sin\beta)$,

$$R_{\max} = \frac{u^2(1-\sin\beta)}{g(1-\sin\beta)(1+\sin\beta)} = \frac{u^2}{g(1+\sin\beta)},$$

where cancelling $1-\sin\beta$ is valid because $\beta < \frac{\pi}{2}$. This is the required distance.

## Checks

- Flat ground, $\beta = 0$: the best angle is $45^\circ$ and the range $u^2/g$, the familiar results.
- As $\beta \to \frac{\pi}{2}$ (a vertical "plane"), the best direction tends to straight up and $R_{\max} \to u^2/(2g)$, the greatest height reached by a ball fired vertically.
- Numerically, with $u = 20\,\mathrm{m\,s^{-1}}$, $g = 9.81\,\mathrm{m\,s^{-2}}$ and $\beta = 30^\circ$, the formula gives $R_{\max} \approx 27.18\,\mathrm{m}$ at elevation $60^\circ$. Stepping the motion in horizontal and vertical coordinates until the ball crosses the plane reproduces $R(\alpha)$ for random $u$, $\beta$, $\alpha$, and a fine search over $\alpha$ puts the maximum at $\frac{\pi}{4} - \frac{\beta}{2}$.

## Common slips

- Mixing up the angle above the horizontal with the angle above the plane; the answer $\frac{\pi}{4} + \frac{\beta}{2}$ is measured from the horizontal.
- Forgetting the component $-g\sin\beta$ along the plane, which gives the flat-ground range formula scaled wrongly.

## Beyond STEP

There is an elegant geometric view: all trajectories with speed $u$ from a point lie inside a "parabola of safety" (their envelope) with focus at the launch point, and the maximum range on any plane is where the plane meets that parabola. The focus–directrix property then gives $R_{\max} = u^2/(g(1+\sin\beta))$ in one line, and explains why the best direction bisects the angle between the plane and the vertical. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good next step for projectile problems treated with vector methods.

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